

Reducing Noise in Noisy Photos

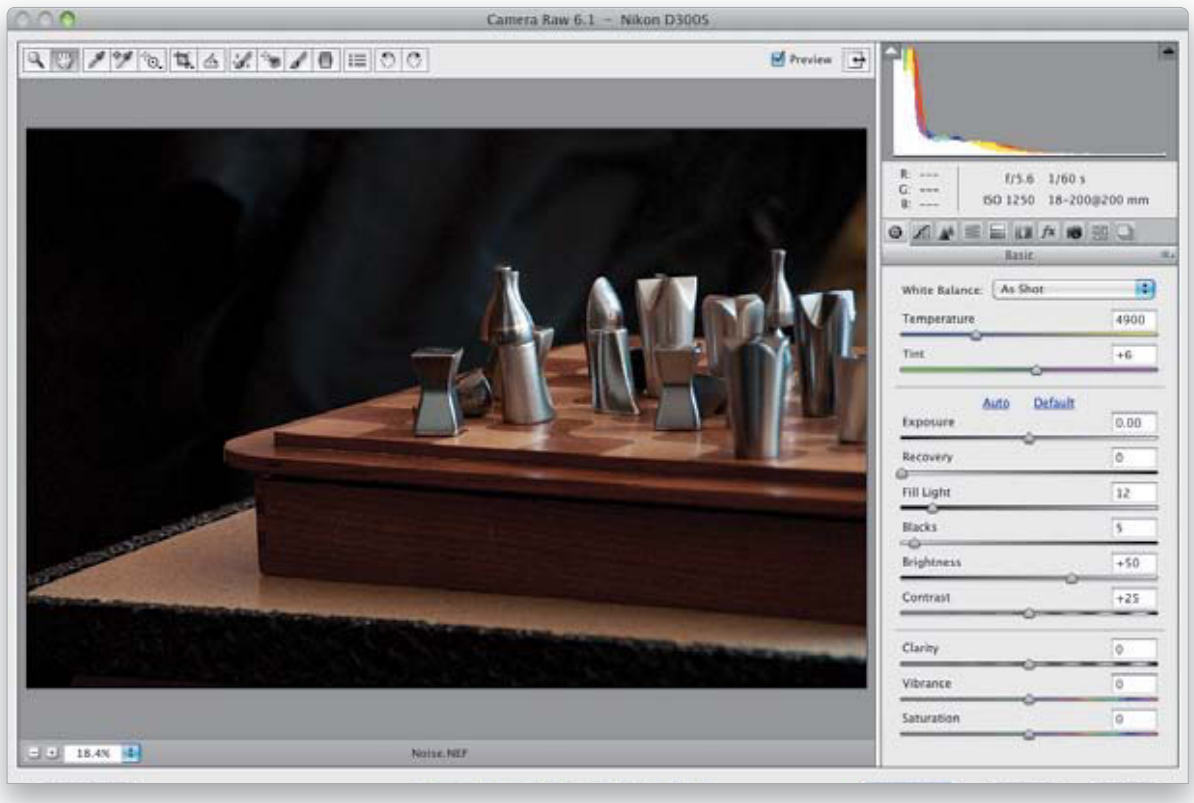
This is, hands down, not only one of the most-requested features by photographers, but one of the best in all of CS5. Now, if you're thinking, "But Scott, haven't Photoshop and Camera Raw both had built-in noise reduction before CS5?" Yes, yes they did. And did it stink? Yes, yes it did. But, does the new noise reduction rock? Oh yeah! What makes it so amazing is that it removes the noise without greatly reducing the sharpness, detail, and color saturation. Plus, it applies the noise reduction to the RAW image itself (unlike most noise plug-ins).

Step One:

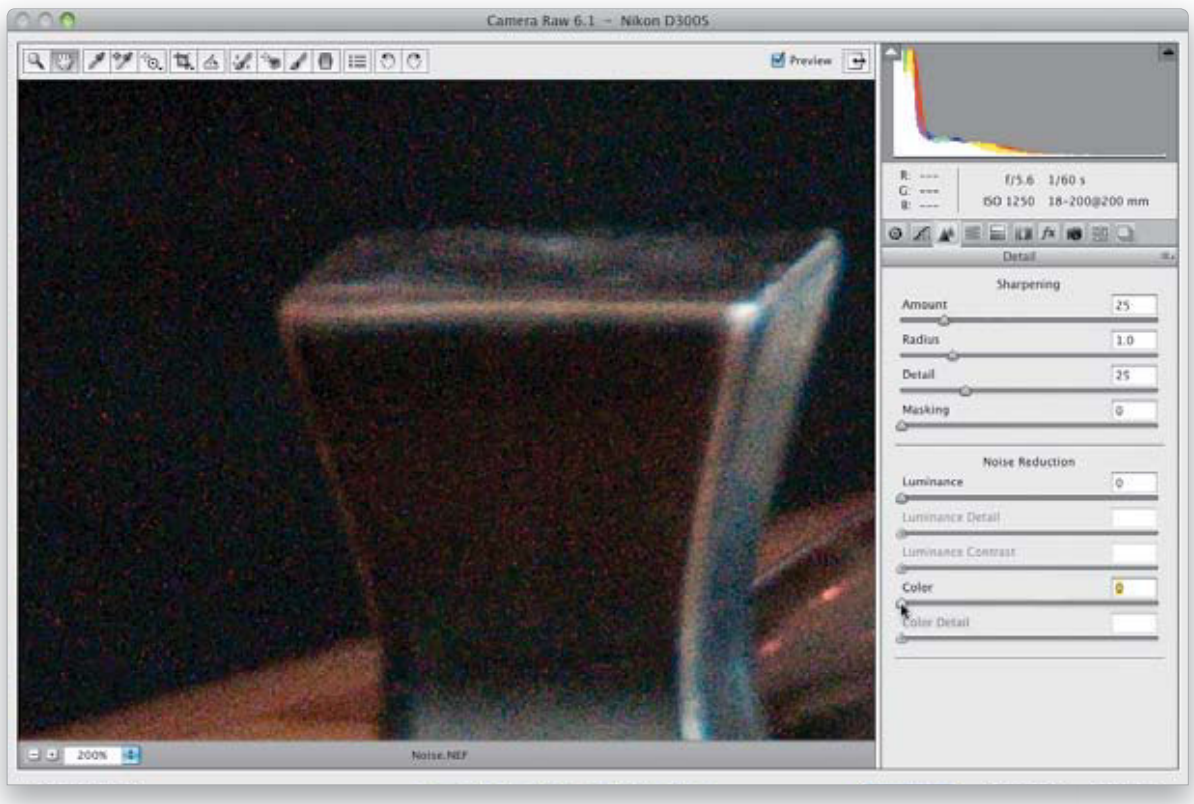
Open your noisy image in Camera Raw (the Noise Reduction feature works best on RAW images, but you can also use it on JPEGs and TIFFs, as well). The image shown here was shot at a high ISO using a Nikon D300S, which, like most cameras in its price range, doesn't do a very good job in low-light situations, so you can expect a lot of color noise (those red, green, and blue spots) and luminance noise (the grainy looking gray spots).

Step Two:

Sometimes it's hard to see the noise until you really zoom in tight, so zoom into at least 100% (here, I zoomed into 200%), and there it is, lurking in the shadows (that's where noise hangs out the most). Click on the Detail icon (it's the third icon from the left at the top of the Panel area) to access the Noise Reduction controls. I usually get rid of the color noise first, because that makes it easier to see the luminance noise (which comes next). Here's a good rule of thumb to go by when removing color noise: start with the Color slider over at 0 (as shown here) and then slowly drag it to the right until the moment the color noise is gone. *Note:* A bit of color noise reduction is automatically applied to RAW images—the Color slider is set to 25. But, for JPEGs or TIFFs, the Color slider is set to 0.

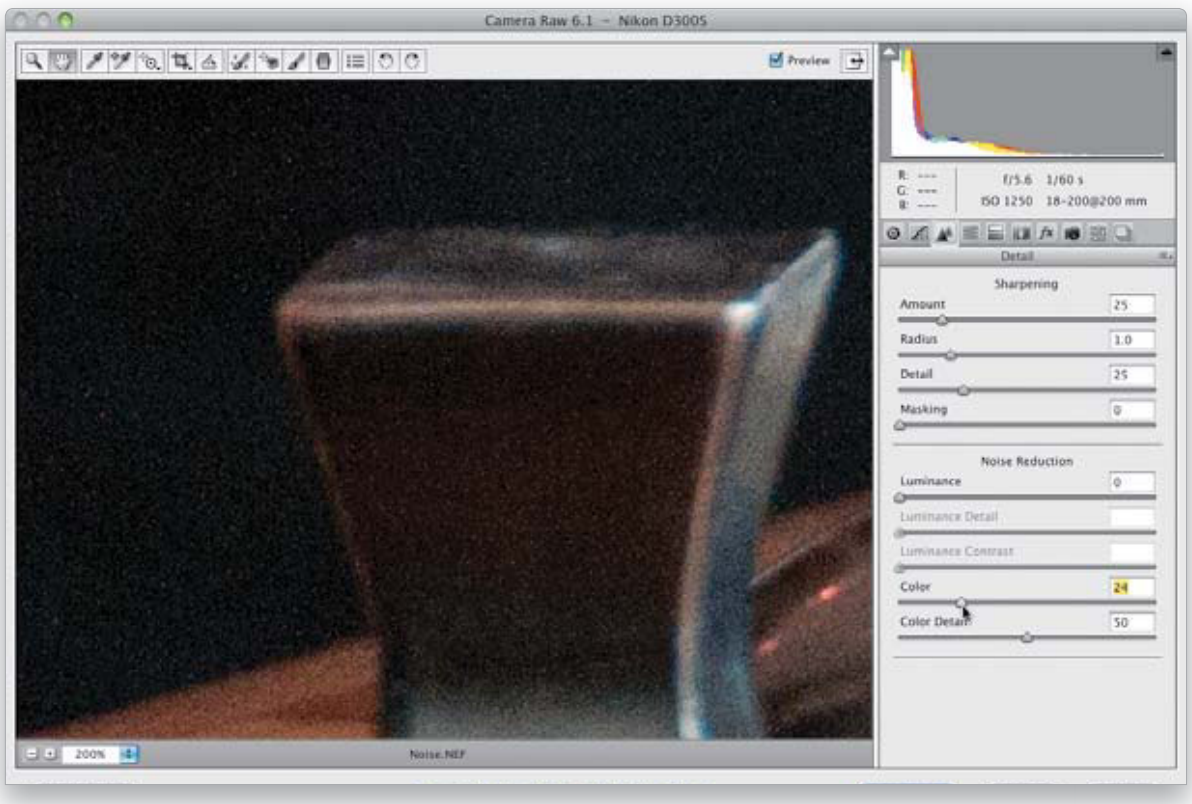


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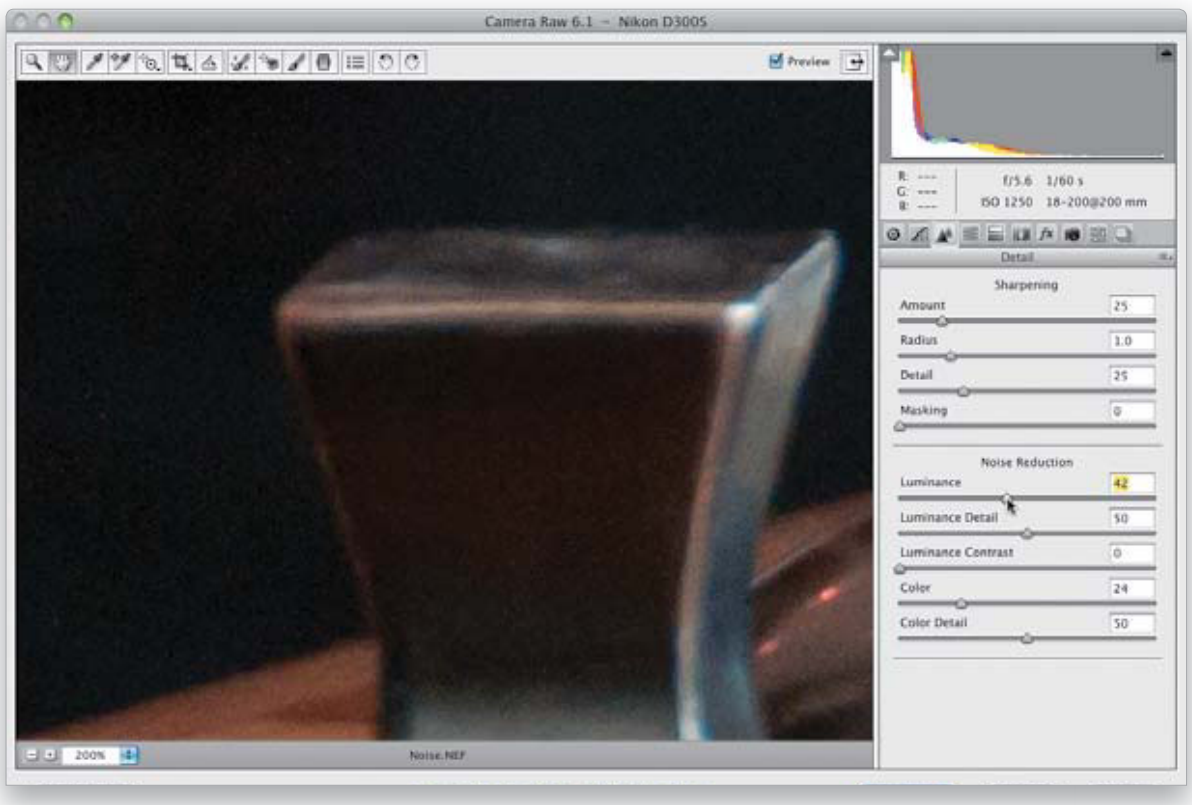
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Step Three:

So, click-and-drag the Color slider to the right, but remember, you'll still see some noise (that's the luminance noise, which we'll deal with next), so what you're looking for here is just for the red, green, and blue color spots to go away. Chances are that you won'zt have to drag very far at all—just until that color noise all turns gray. If you have to push the Color slider pretty far to the right, you might start to lose some detail, and in that case, you can drag the Color Detail slider to right a bit, though honestly, I rarely have to do this for color noise.



Step Four:

Now that the color noise is gone, all that's left is the luminance noise, and you'll want to use a similar process: just drag the Luminance slider to the right, and keep dragging until the visible noise disappears (as seen here). You'll generally have to drag this one farther to the right than you did with the Color slider, but that's normal. There are two things that tend to happen when you have to push this slider really far to the right: you lose sharpness (detail) and contrast. Just increase the Luminance Detail slider if things start to get too soft (but I tend not to drag this one too far), and if things start looking flat, add the missing contrast back in using the Luminance Contrast slider (I don't mind cranking this one up a bit, except when I'm working on a portrait, because the flesh tones start to look icky). You probably won't have to touch either one all that often, but it's nice to know they're there if you need them.

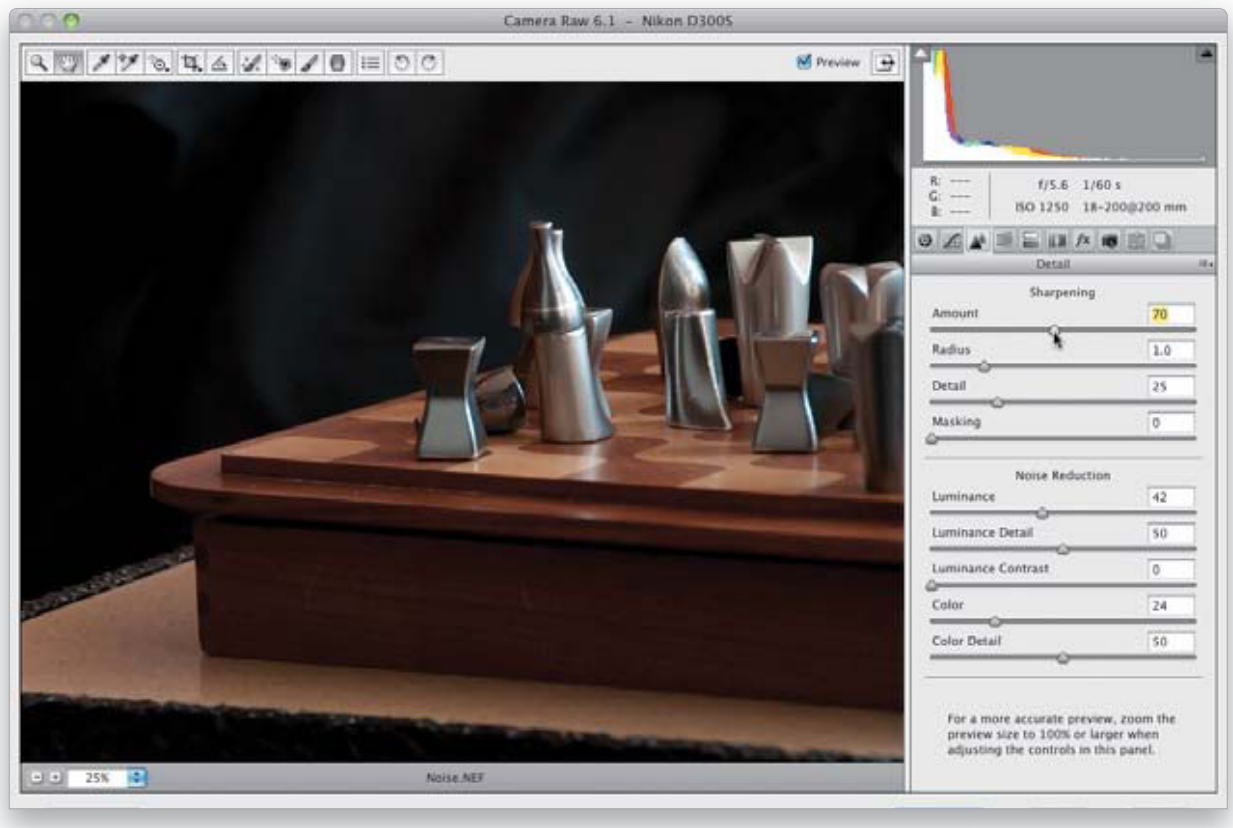
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Step Five:

Rather than increasing the Luminance Detail a bunch, I generally bump up the Sharpening Amount at the top of the Detail panel (as shown here), which really helps to bring some of the original sharpness and detail back. Here's the final image, zoomed back out, and you can see the noise has been pretty much eliminated, but even with the default settings (if you're fixing a RAW image), you're usually able to keep a lot of the original sharpness and detail. A zoomed-in before/after of the noise reduction we applied here is shown below.





Before



After

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